

NE 511: Multi-Physics of Nuclear Reactors

Exam 1 : Study Guide and Examples

Exam 1 will cover :

Fundamentals of Reactor Multi-Physics

Short-Time Multi-Physics Phenomena in Reactor Core

Exam 1 will consist of ten (10) questions (examples are following) and one (1) extra credit question.

Exam 1 is an open book take home exam. However, it is a timed exam - you will have 100 minutes to open it, work on it, scan and submit it. It will be administrated via Moodle.

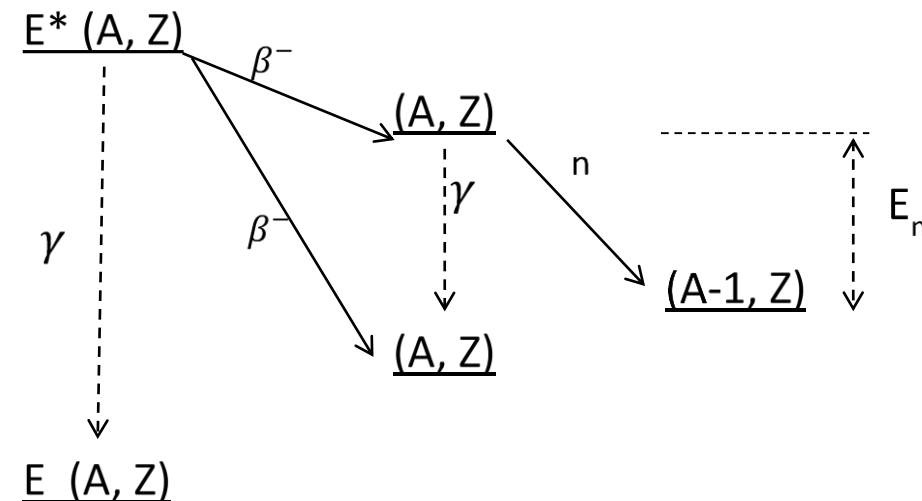
Delayed Neutrons - Review

What are delayed neutron precursors and delayed neutron emitters?

Precursor nuclei are some of the fission product nuclei that will decay into emitter nuclei with beta emission.

Emitter nuclei are the daughter nuclei that emit delayed neutrons.

Sketch the typical level scheme of precursor-emitter-daughter nuclei that leads to the production of delayed neutrons?



Delayed Neutrons

What is the known or estimated approximate number of possible precursors?

Forty (40)

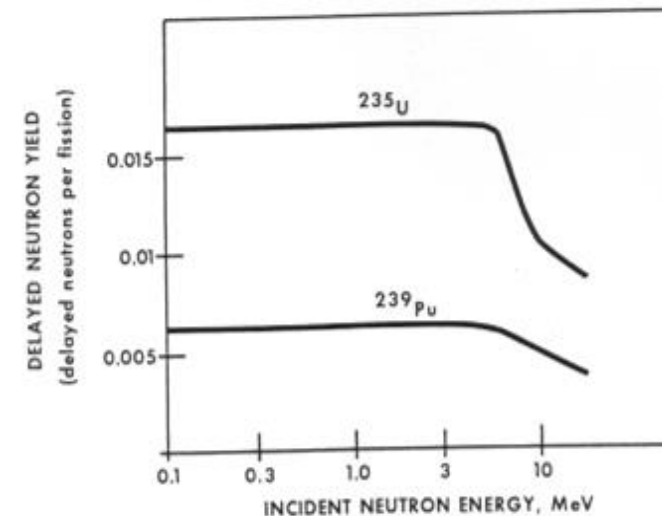
Name the quantities used to describe the physics of precursors and delayed neutrons.

Delayed neutron yield, ν_d

Delayed neutron decay constants λ_d

Delayed neutron spectra, E_d

Sketch the typical E dependence of $\nu_d(E)$ for ^{235}U and ^{239}Pu .



How many delayed neutron groups (families) are generally used per fissioning isotope?

Six (6)

Delayed Neutrons

What is the disadvantage of using isotope-dependent decay constants?

1. *The lifetimes of many precursors are not known accurately enough*
2. *Very complicated theoretical formulation of the kinetics problem leading to an impractically lengthy set of differential equations*
3. *Several precursors are themselves products of beta decays - multiple beta decay schemes*

Give the simplest form of the balance equation for the total number of neutrons in a reactor, $n(t)$. Use one-group constants and disregard delayed neutrons.

$$\frac{dn(t)}{dt} = \nu \Sigma_f \hat{\Phi}(t) - (\Sigma_a + DB^2) \hat{\Phi}(t)$$

Point Kinetics Equations

Give the balance equation of the delayed neutron precursors in its most direct form.

$$\frac{d\hat{C}_k}{dt} = -\lambda_k \hat{C}_k + \nu_{dk} \sum_f \hat{\Phi}, \quad k = 1, \dots, 6$$

In which way is the source term in this equation often rewritten?

$$\hat{S}_{pk} = \beta_k \nu \sum_f \bar{v} n \qquad \hat{S}_{pk} = \frac{\beta_k}{\Lambda} n$$

Point Kinetics Equations

Derive the simplest forms of the point kinetics equations based on the concepts of $\hat{\Phi}, \rho, \beta$ and, Λ in the ζ notation.

(self preparation)

$$\Phi(r, E, t) = p(t)\psi(r, E)$$

...

$$\frac{dp}{dt} = \frac{\rho - \beta}{\Lambda} p + \frac{1}{\Lambda} \sum_k \lambda_k \zeta_k + \frac{1}{\Lambda} s(t)$$

$$\frac{d\zeta_k}{dt} = -\lambda_k \zeta_k + \beta_k p(t)$$

Point Kinetics Equations

Give the space-, energy-, and time-dependent diffusion equation in operator form.

$$\frac{1}{v} \frac{d\Phi(r, E, t)}{dt} = (F_p - M)\Phi(r, E, t) + S_d(r, E, t) + S(r, E, t)$$

Define the operators.

$$F_p \Phi = \sum_i \chi_{pi}(E) \int_{E'} v_{pi}(E') \Sigma_{fi}(r, E', t) \Phi(r, E', t) dE'$$

$$M\Phi = -\nabla \cdot D(r, E, t) \nabla \Phi(r, E, t) + \Sigma_t(r, E, t) \Phi(r, E, t) - \int_{E'} \Sigma_s(r, E' \rightarrow E, t) \Phi(r, E', t) dE'$$

$$S_d(r, E, t) = \sum_k \lambda_k C_k(r, t) \chi_{dk}(E)$$

Point Kinetics Equations

What does the assumption of “stationary fuel” mean?

No physical transport of precursors - the precursors can be assumed to decay at the location where they are produced

Give two examples where this assumption is violated.

Severe accidents – fuel melting or dislocation
Molten salt reactors

Point Kinetics Equations

Separate the flux into amplitude and shape functions.

$$\Phi(r, E, t) = p(t)\psi(r, E)$$

Give the time- and energy-dependent diffusion equation (use operators as far as reasonable).

$$\begin{aligned} \frac{1}{v} \frac{d\Phi(r, E, t)}{dt} - \nabla \cdot D(r, E) \nabla \Phi(r, E, t) + \Sigma_r(r, E) \Phi(r, E, t) = & \int_0^\infty dE' \Sigma_s(r, E' \rightarrow E) \Phi(r, E', t) \\ & + \chi_p(E) \int_0^\infty dE' v_p \Sigma_f(r, E') \Phi(r, E', t) + \chi_d(E) \sum_l \lambda_l C_l(r, t) + S(r, E, t) \end{aligned}$$

Point Kinetics Equations

Outline the derivation of the one-group point kinetics equations and give the result.

Step 1: Substitute the flux, $\Phi(\mathbf{r}, \mathbf{E}, t) = p(t) \psi(\mathbf{r}, \mathbf{E})$, into the neutron diffusion equation.

Step 2: Integrate over energy to obtain the one-group equation.

Step 3: Integrate over the volume of the reactor and introduce the buckling.

Step 4: Define ρ , β , Λ , and ζ .

Step 5: Consider the precursor balance equations

$$\frac{dp}{dt} = \frac{\rho - \beta}{\Lambda} p + \frac{1}{\Lambda} \sum_k \lambda_k \zeta_k + \frac{1}{\Lambda} s(t)$$

$$\frac{dp}{dt} = \frac{\rho - \beta}{\Lambda} p + \sum_k \lambda_k c_k + s_c$$

$$\frac{d\zeta_k}{dt} = -\lambda_k \zeta_k + \beta_k p(t)$$

$$\frac{dc_k}{dt} = -\lambda_k c_k + \frac{\beta_k}{\Lambda} p$$

Point Kinetics Equations

Factorize the flux and give the rationale for the factorization

The time-, space-, and energy-dependent neutron flux is factorized into a purely time-dependent amplitude function, $p(t)$, and a space-, energy-, and time-dependent shape function, $\psi(r,E,t)$:

$$\phi(\mathbf{r}, E, t) = p(t) \cdot \psi(\mathbf{r}, E, t)$$

Point Kinetics Equations

Factorize the flux and give the rationale for the factorization

The flux factorization is particularly advantageous if the flux variation with time consists primarily of flux amplitude changes with comparatively small time variations of the shape function

The neutronics equation is multiplied by a weight function, $w(r,E)$, prior to integration with respect to space and energy

A second equation is required to make the factorization unique - it shifts the major time dependence into the amplitude function by constraining the time variation of the “magnitude” of the shape function

$$\int_V \int_0^\infty \frac{w(\mathbf{r}, E) \psi(\mathbf{r}, E, t)}{\nu(E)} dE dV = \text{constant}$$

Point Kinetics Equations

Which weighting function is commonly chosen?

The initial adjoint flux: $\phi_0^(\mathbf{r}, E)$*

Give the rationale for this choice.

We need a weight function that reduces the effect of the shape function inaccuracies on the reactivity (the solution of the point kinetics equation is particularly sensitive to an error in the reactivity)

$$w(\mathbf{r}, E) = \phi_0^*(\mathbf{r}, E) \longrightarrow \int_V \int_0^\infty \frac{\phi_0^*(\mathbf{r}, E) \psi(\mathbf{r}, E, t)}{\nu(E)} dE dV = K_0$$

Point Kinetics Equations

What does “exact” mean in the context of exact point kinetics equations?

“Exact” means that a given time-, space-, and energy-dependent neutronics model is condensed or lumped into the form of the point kinetics equations without applying simplifying approximations

Point Kinetics Equations

What is dynamic reactivity and why is it different from the static reactivity?

The dynamic reactivity is formed with the time-dependent flux, $\Phi(r, E, T)$, as it physically appears during a transient

$$\rho(t) = \frac{1}{F(t)} (w(r, E), [\Delta F - \Delta M] \Psi)$$

$$\rho(t) = \rho_0 + \delta \rho(t)$$

$$\rho_0 = \rho_0^{\text{st}}$$

$$\delta \rho(t) = \frac{1}{F(t)} (w(r, E), [F - M] \Psi)$$

Point Kinetics Equations

What is the major difference between effective β and corresponding quantity of one-group point kinetics?

These differences result from the energy as well as the space dependence of the adjoint flux:

- 1. The energy dependence of the adjoint flux provides the proper importance weighting for the emission spectra of the delayed neutrons*
- 2. The space dependence weighs the spatial distribution of the delayed neutron emission*

The Point Reactor Model

What is the basic assumption in the point reactor model?

The major simplification consists of neglecting the time dependence of the flux shape

The initial flux shape is used in forming the integral kinetics parameters

$$\psi(\mathbf{r}, E, t) \cong \phi_0(\mathbf{r}, E)$$

The Point Reactor Model

What are the two kinds of flux shape variations commonly accounted for in the point reactor model?

Point kinetics is a good approximation when the flux shape is largely invariable during a transient:

- 1. tightly coupled small reactors*
- 2. transients in large reactors if they involve fairly uniform perturbations*

Feedback Effects

Give an example of a transient for the application of energy, temperature, and power coefficients, respectively.

Energy coefficient - rapid transients described by the power change (rod ejection)

Temperature coefficient – Rod Withdrawal at Power

Power coefficient - rapid transients described by change in the flux amplitude (in-phase instability oscillation)

Feedback Effects

Give the formula for the adiabatic feedback model for the entire and the incremental energy respectively

Give a typical application for each of the two models

$$\delta T(t) = C_{QT} \int_0^t P(t') dt' \begin{cases} \text{completely} \\ \text{adiabatic fuel} \end{cases} \quad \text{Atomic Bomb}$$

$$\delta T(t) = C_{QT} \int_0^t [P(t') - P_0] dt' \begin{cases} \text{adiabatic fuel for} \\ \text{power increment} \end{cases}$$

Rod Ejection at Hot Zero Power

Feedback Effects

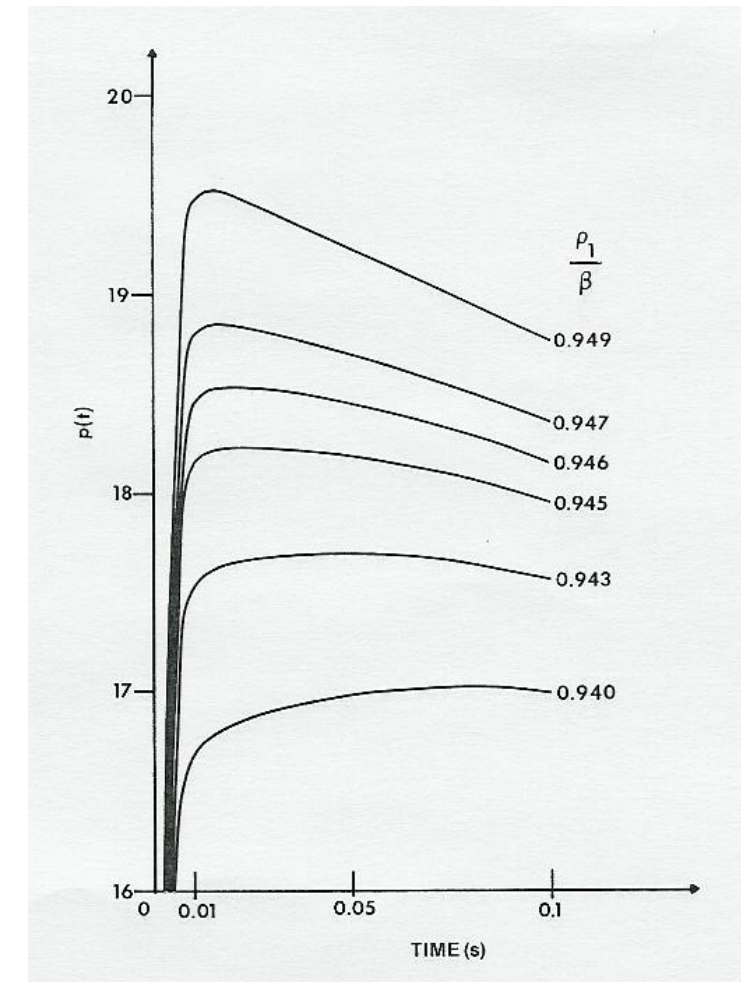
Describe the transient at small times (including the two competing effects). Which approximation allows an accurate quantitative evaluation of those effects?

$$\underbrace{\Delta \dot{p}}_{=0} = \left\{ \underbrace{\rho_1 - \beta}_{\text{reactivity}} + \underbrace{\gamma \int_0^t [p(t') - p_0] dt'}_{\text{feedback}} \right\} p + \underbrace{\sum_k \lambda_k \zeta_k}_{\text{delayed neutrons}}$$

The total reactivity slope is:

$$\begin{aligned} p' &= p'_{\text{del}} + p'_{\text{fb}} \\ &= \bar{\lambda} \frac{\beta}{\beta - \rho_1} (p^0 - p_0) + \frac{\gamma}{\beta} \frac{p^0}{p_0} (p^0 - p_0) p^0 \\ &= \left[\bar{\lambda} + \frac{\gamma}{\beta} p^0 \right] \frac{p^0}{p_0} (p^0 - p_0) \end{aligned}$$

PJA + one-group kinetics + adiabatic fuel



Feedback Effects

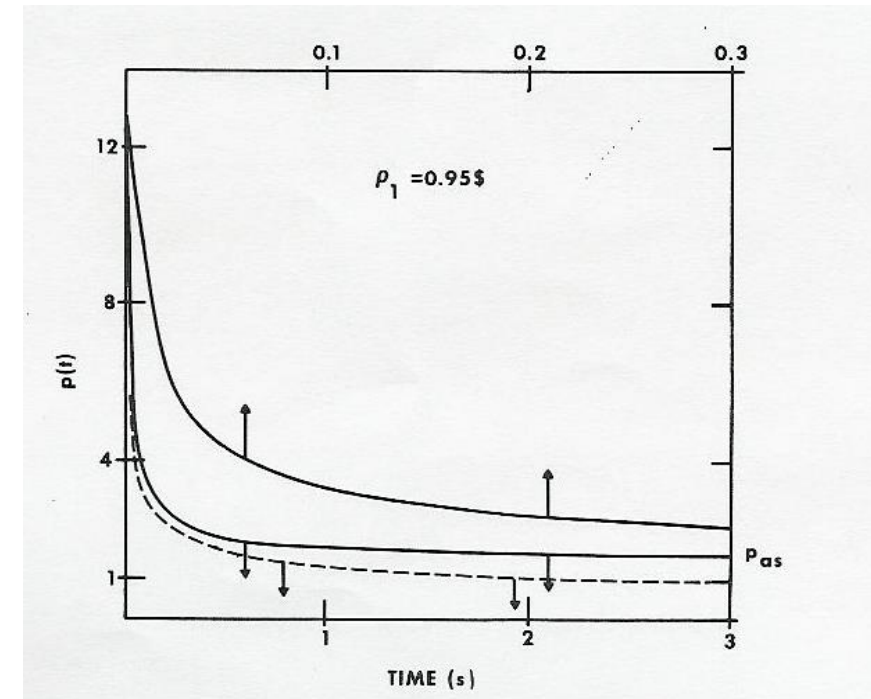
Find the value of the asymptotic flux (derived from the assumption that it is constant)

a) without heat transfer

$$p_{as} = p_0$$

b) with heat transfer described by a single exponential function

$$p_{as} = p_0 - \rho_1 \frac{\lambda_H}{\gamma}$$



Feedback Effects

Why is the feedback reactivity approximated by neglecting the stationary heat removal in the analytical treatment of step-induced super-prompt critical transients?

- The stationary cooling is neglected, (the term “ $-p_0$ ” is omitted in the integrand) → physically not well justified in most transients, but it greatly simplifies the mathematical treatment
- The stationary cooling during a transient is equivalent to a positive ramp reactivity insertion (and therefore cannot be neglected),

$$\gamma \int_0^t (-p_0) dt' = -\gamma p_0 t$$

Feedback Effects

For step-induced superprompt-critical transients, give the differential equation in the prompt kinetics approximation, including the initial condition.

The differential equation and the initial condition in the prompt kinetics approximation are given by

$$\dot{p} = \frac{\rho - \beta}{\Lambda} p$$

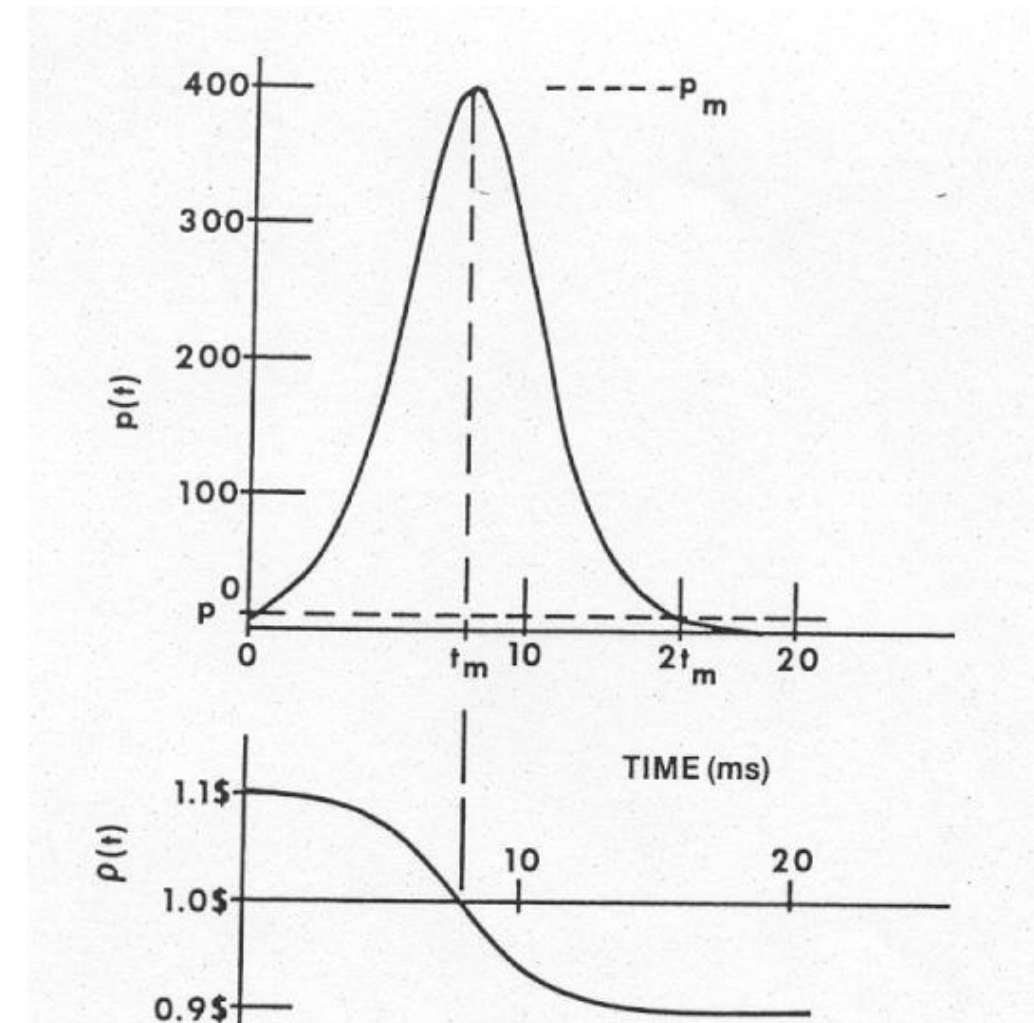
with

$$p(0) = p^0 = \frac{\rho_1}{\rho_1 - \beta} p_0$$

Feedback Effects

For step-induced super-prompt critical transients, find the reactivity at the maximum of the transient

$$\left. \begin{array}{l} \dot{p} = 0 \\ \rho - \beta = \rho_p \\ \rho_p(t_m) = 0 \end{array} \right\} \rho(t_m) = \beta$$



Feedback Effects

For step-induced super-prompt critical transients, find the energy release up to the maximum of the transient

$$\rho(t) = \rho_1 + \gamma \int_0^t p(t') dt'$$

$$-\gamma \int_0^{t_m} p(t') dt' = \rho_1 - \beta$$



$$\Delta Q_m = \int_0^{t_m} P(t') dt' = -\frac{\rho_{p1}}{\gamma_e}$$

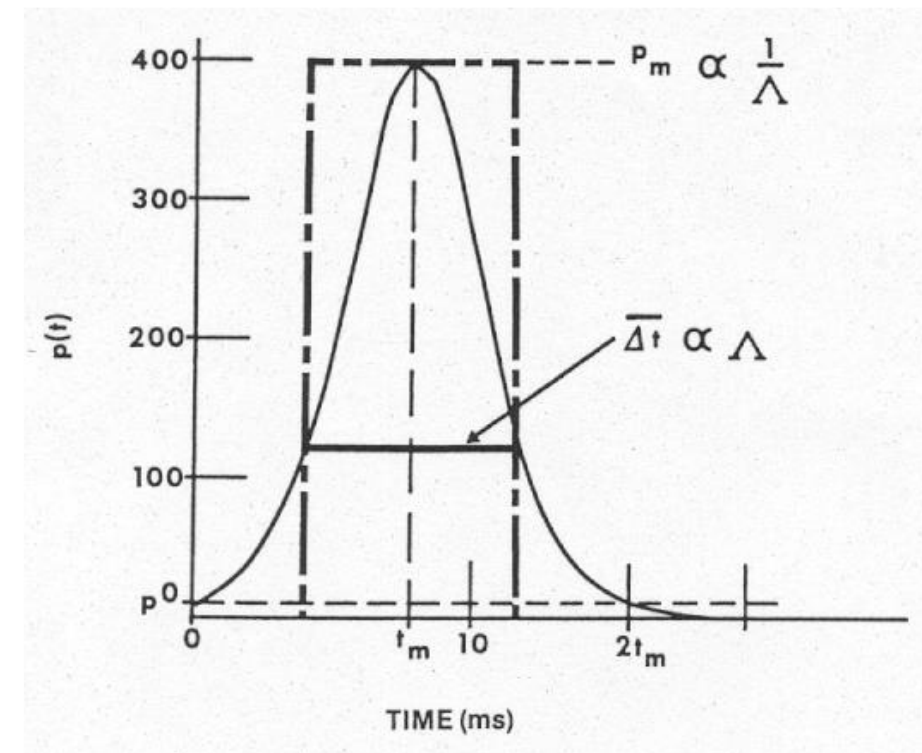
Feedback Effects

Define the width of the flux pulse. Give the main factors of the formula for the width of the flux pulse and show why the energy release is independent of Λ

The simplest formula for the "width" of the flux pulse (or burst) is obtained when the width $\bar{\Delta}t$ is defined by the flux integral divided by the maximum value:

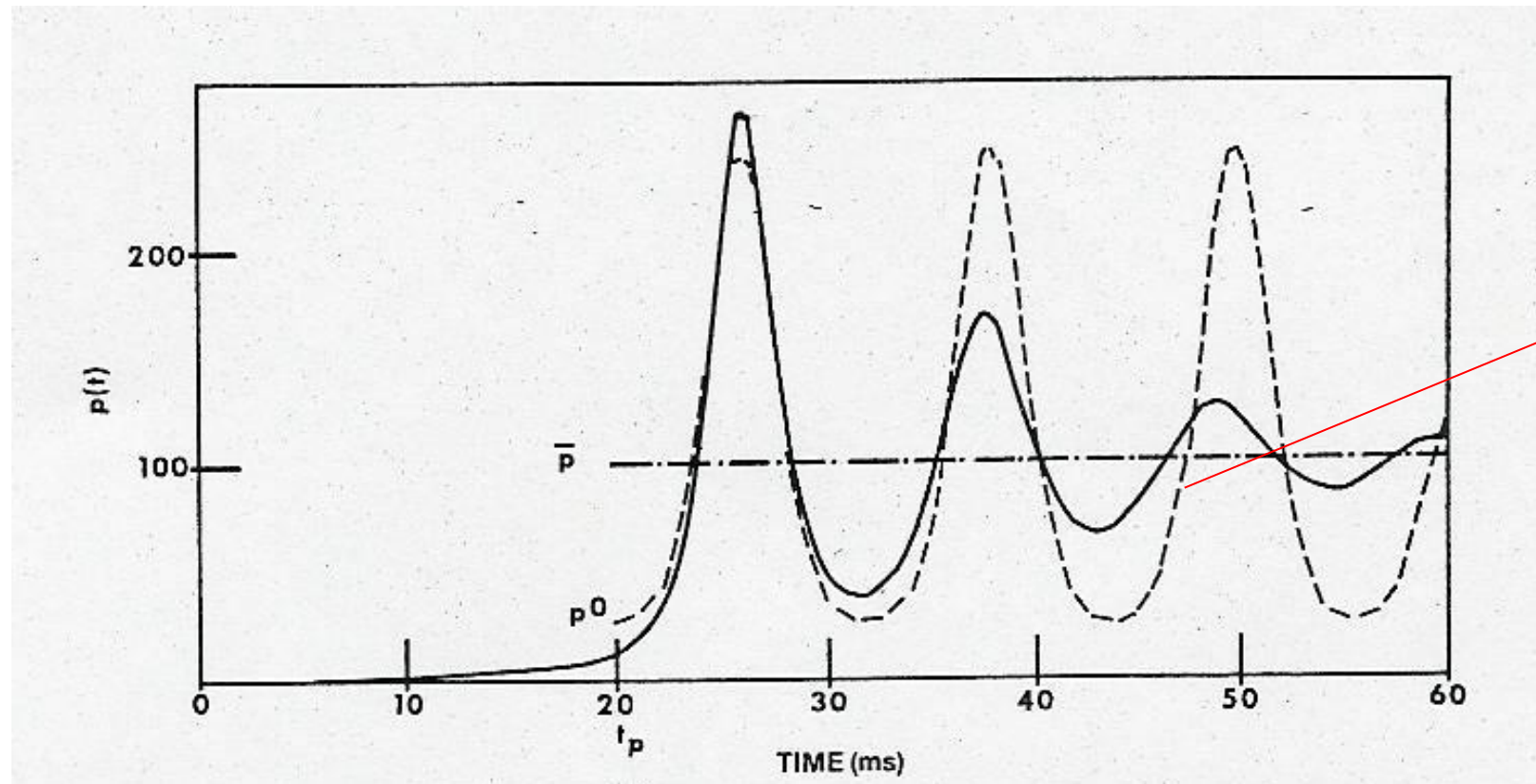
$$\bar{\Delta}t = \frac{4\Lambda}{\rho_{p1}} \left(1 - \frac{p^0}{p_m} \right)$$

$$\bar{\Delta}t = \frac{4\Lambda}{\rho_{p1}} \quad \text{for } p_m \gg p^0$$



Feedback Effects

Discuss the principal effect of the delayed neutron on the transient as treated in the prompt kinetics approximation

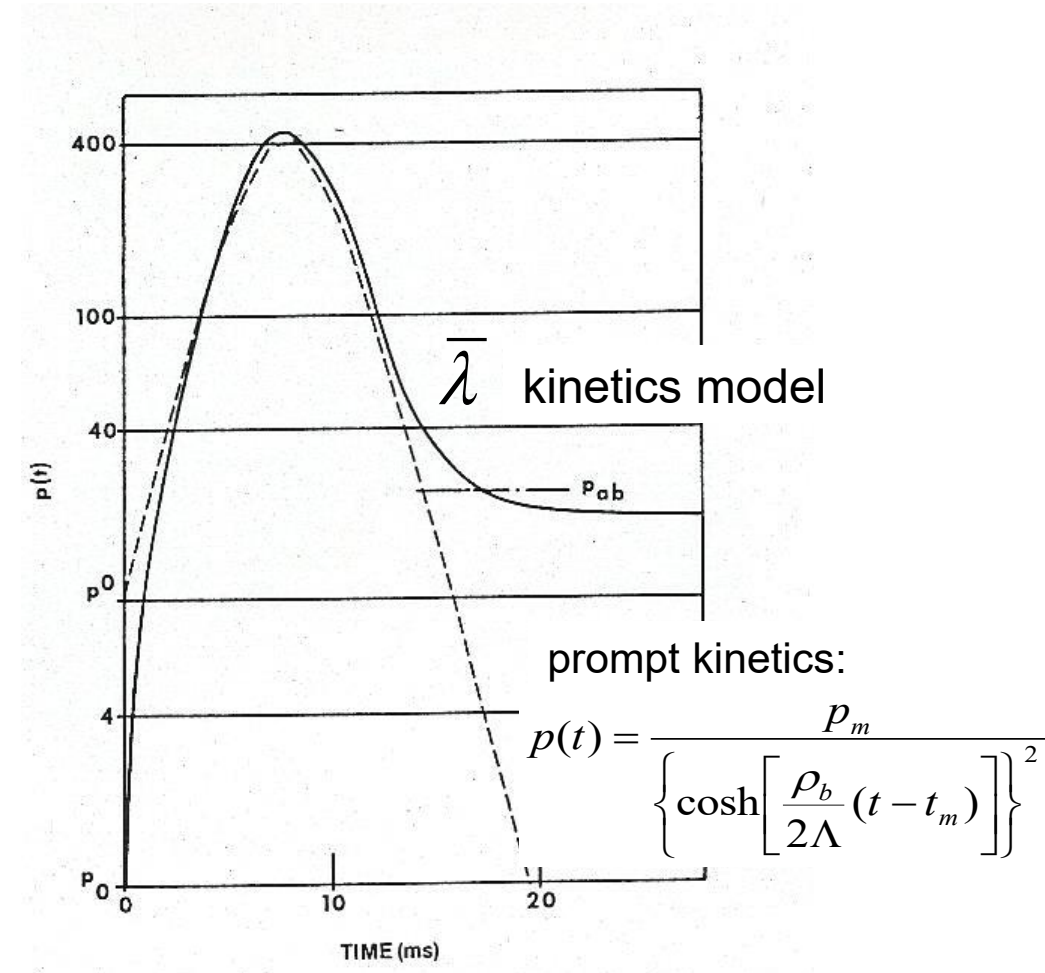
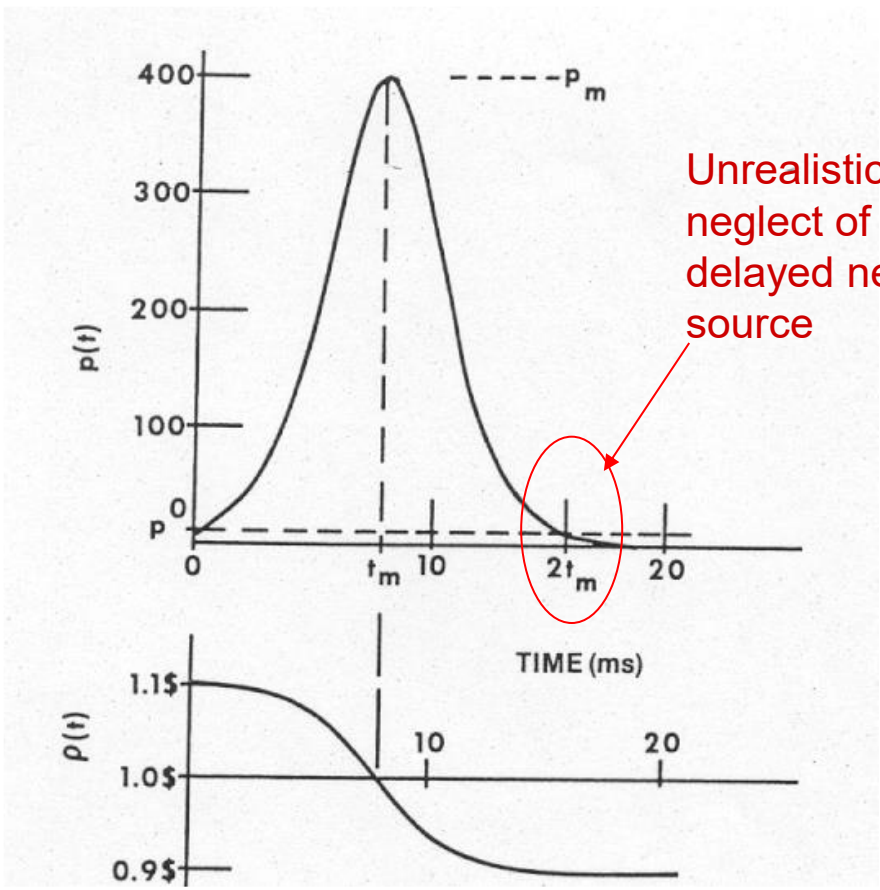


Effect of Delayed neutrons

The flux transient induced by a reactivity ramp in the PKA (dashed line) and the actual solution (solid line) with a temperature-independent energy coefficient

Feedback Effects

Discuss the applicability of the prompt kinetics approximation (for $\rho > \beta$ and for $\rho < \beta$)



Super prompt critical transient ($\rho_1 = 1.1\$, p_0 = 0.1p_n$) with reactivity feedback

Feedback Effects

Describe the "initial" condition as it is used in the prompt kinetics approximation for super-prompt critical transients induced by reactivity ramps

The PKA for reactivity ramps is applied (as for the step-induced transients) only after the time, t_p , when the reactor became super-prompt critical:

$$\dot{p}(\tau) = \frac{\rho_p(\tau)}{\Lambda} p(\tau) \quad \text{with} \quad \tau = t - t_p$$

which is solved with $p(0) = p^0$, and $\rho_p(0) = 0$ for $\tau = 0$

Feedback Effects

Give the differential equation, the initial condition, and the reactivity including prompt feedback

The prompt reactivity, with the adiabatic approximation applied to the energy above p_0 is given by

$$\rho_p(\tau) = a\tau + \gamma \int_0^\tau [p(\tau') - p_0] d\tau'$$

or

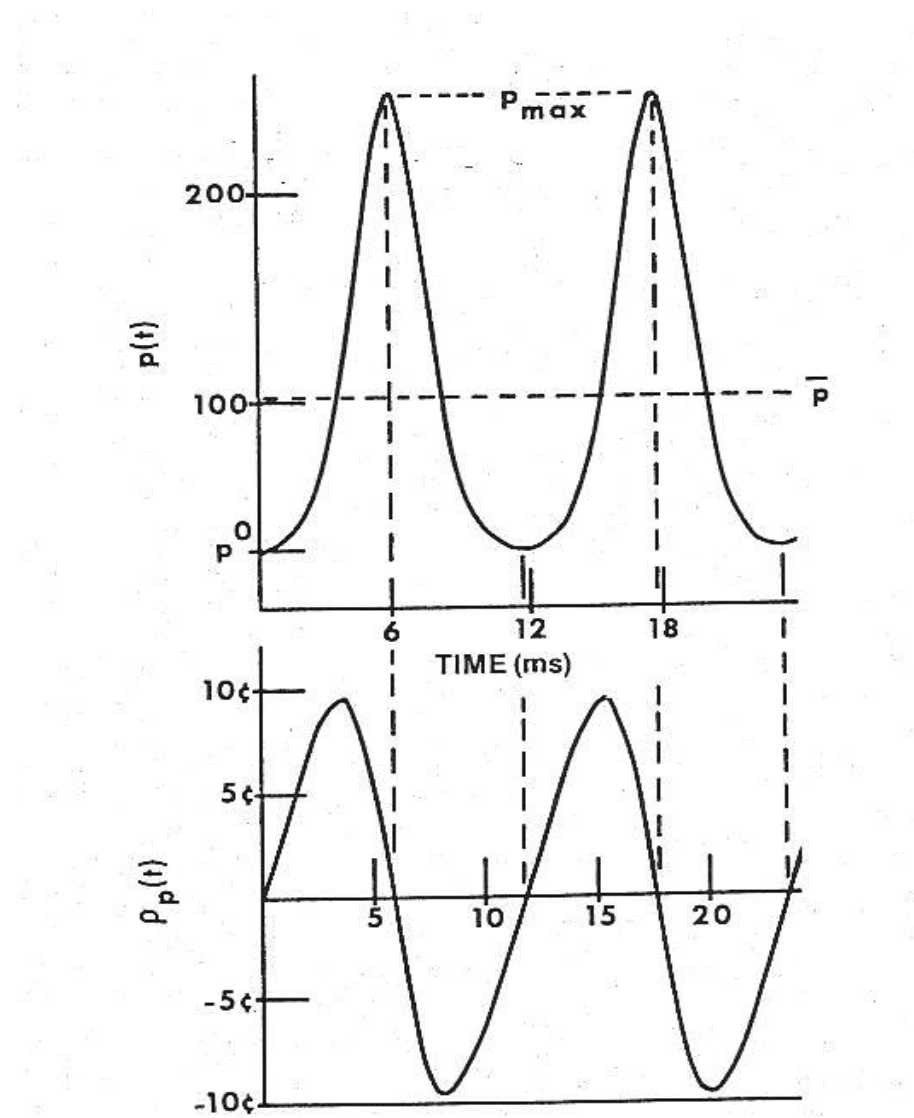
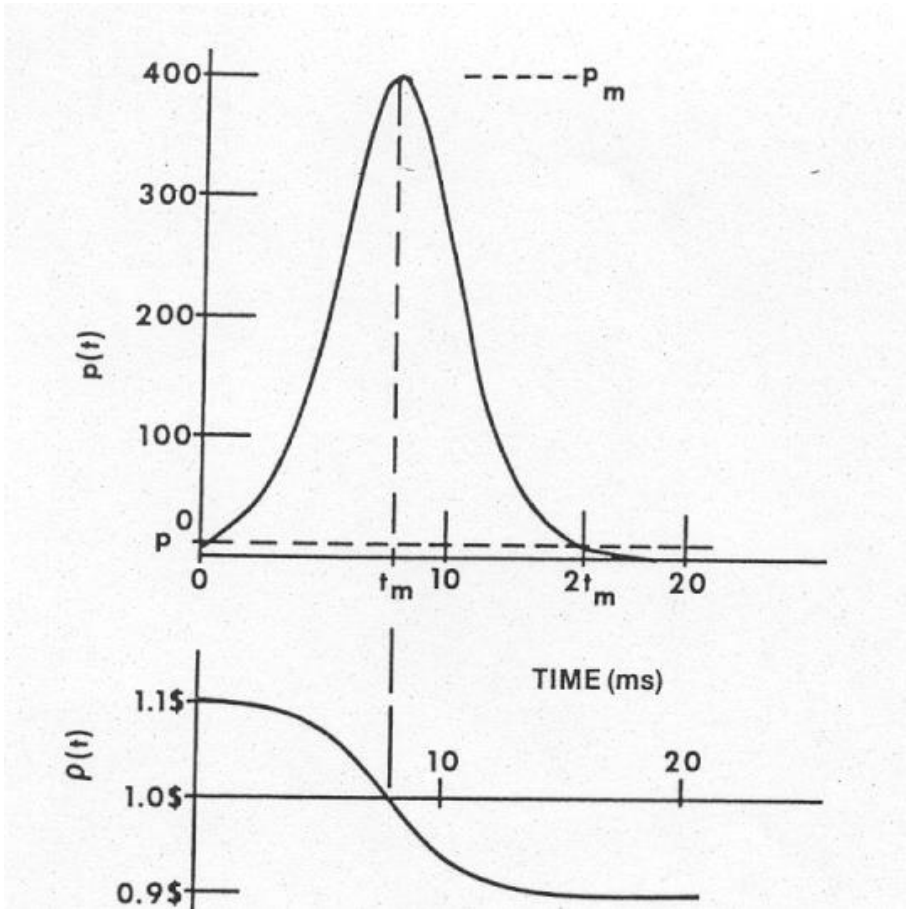
$$\rho_p(\tau) = a'\tau + \gamma \int_0^\tau p(\tau') d\tau'$$

with

$$a' = a + \gamma p_0$$

Feedback Effects

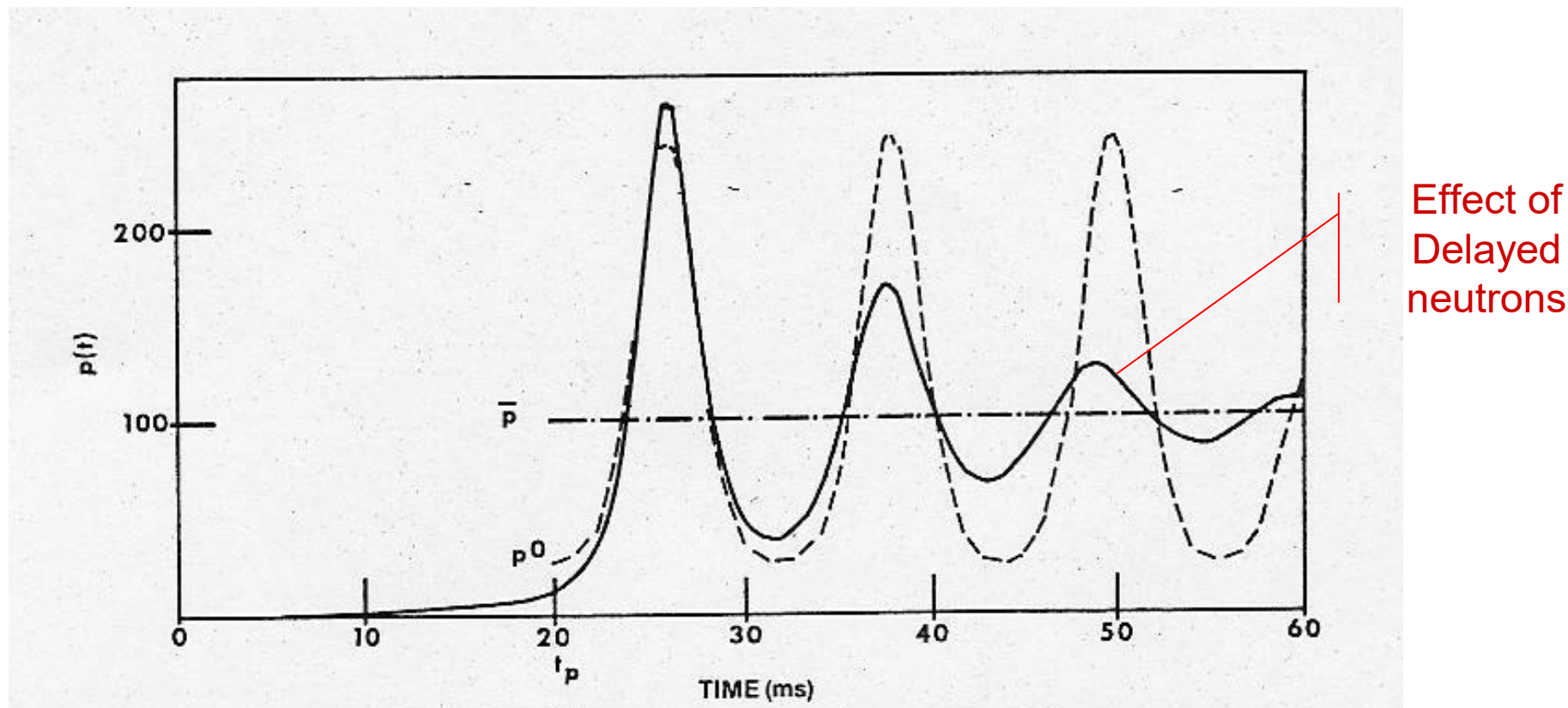
Sketch qualitatively the time dependence of the reactivity and the flux in the prompt kinetics approximation and relate key points



Feedback Effects

Describe qualitatively the impact of the increase of the delayed neutron source on the first post-burst minimum.

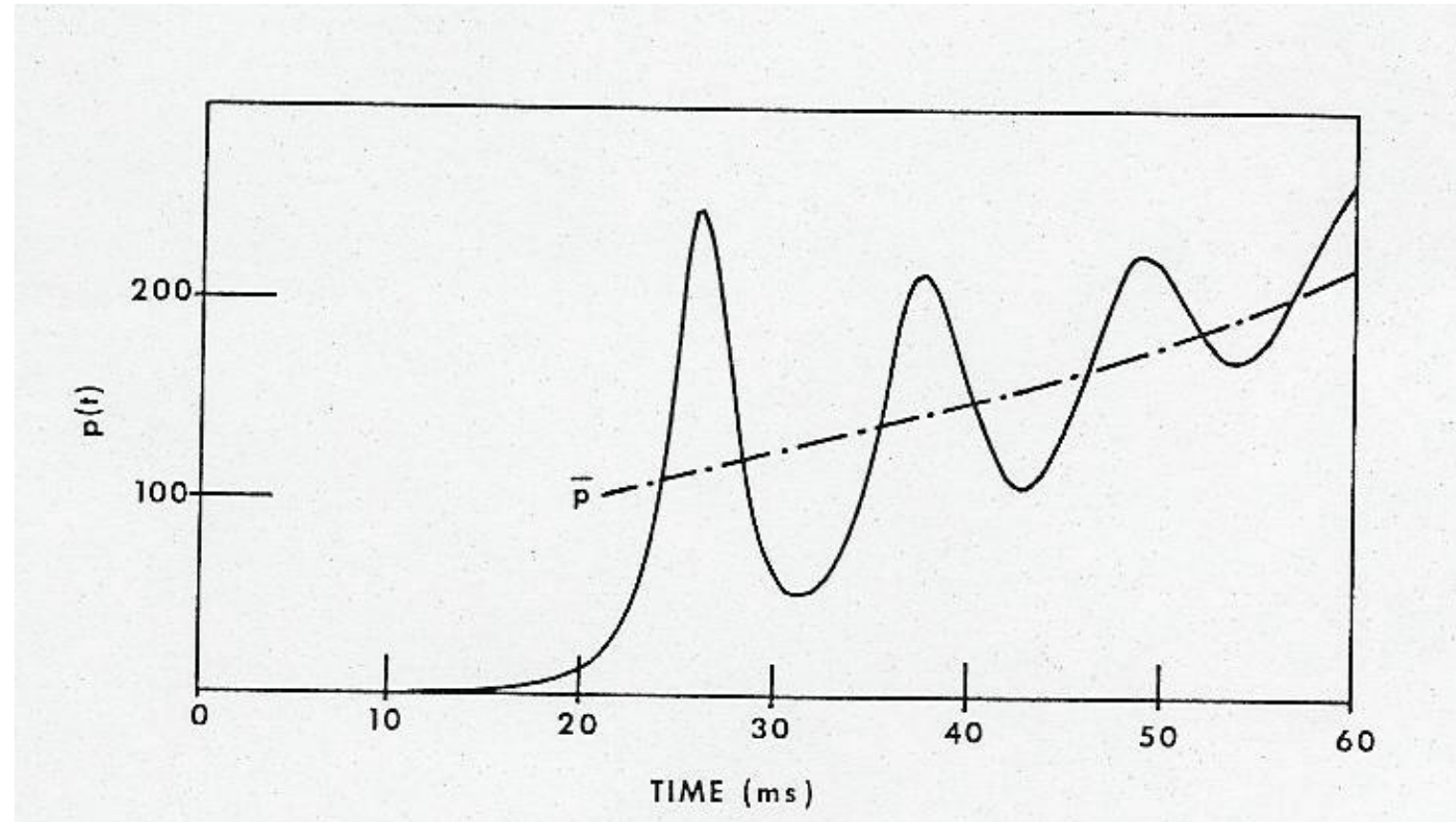
Explain the damping effect of the delayed neutrons.



The flux transient induced by a reactivity ramp in the PKA (dashed line) and the actual solution (solid line) with a temperature-independent energy coefficient

Feedback Effects

Suppose $\gamma(T)$ decreased with T ; describe the resulting two changes in the transient



The flux transient induced by a reactivity ramp - actual solution with a temperature-dependent energy coefficient